AI Automation Opportunity Report 4

Prepared for: Jeffway Construction Ltd **Prepared by:** Cybix **Date:** 23 March 2026

---

1. Overview

This report provides a definitive, fully-scoped breakdown of the AI-powered automation opportunity for Jeffway Construction Ltd, specifically focusing on the electrical estimating, commercial tracking, and labour processing lifecycle. The purpose of this document is to detail, in black and white, every step, trigger, and action that will be built, ensuring complete clarity on the solution being delivered. This automation is designed to act as a digital estimator and commercial assistant within your existing infrastructure, seamlessly connecting your tender documents, supplier quotes, EQUE2 payroll system, Monday.com pipeline, and financial tracking spreadsheets.

The primary business objective is to eliminate the significant administrative burden which currently consumes the commercial and estimating teams, introduces pricing risks, and creates blind spots in project profitability. By automating the repetitive, rules-based tasks of drawing takeoffs from complex 46-page architectural packs, multi-supplier price comparisons, PDF timesheet processing, and material invoice checking, we will unlock substantial value, improve estimating accuracy, and ensure that every project's profitability is protected from the tender stage right through to final completion.

Mini Report Note

I will add this is just a mini report, as there are a couple of automation workflows focused on the commercial and estimating process, and we will do the same kind of style report as this for the other sessions we have. In the full report, we'll have a look at your competitive market, different tools the team can use internally, and lots more, but this should give you a bit of a flavour for this smaller automation process.

---

2. The Problem: Current Process & Inefficiencies

Jeffway Construction's current process for electrical estimating and commercial tracking relies heavily on manual data extraction and spreadsheet management. This creates a series of tangible business challenges that directly impact commercial capacity and margin protection.

The core of the problem begins at the tender stage. When a tender pack estimators must manually review extensive architectural drawing packs (often 40+ pages per house type), counting every single socket, light fitting, and cable run by hand. This data is then manually transcribed into complex, multi-tab spreadsheets separating 1st Fix, 2nd Fix, and Communals. The process of obtaining supplier quotes involves sending out lists to up to six different suppliers (including Templegate, Medlock, and KEW), waiting for returns in varying formats, and manually comparing line items. Once a project is won, the burden shifts to the commercial team, who must manually process handwritten PDF timesheets from

operatives, check them against allowed budgets per fix stage, enter the data into EQUE2, and painstakingly review hundreds of supplier invoice line items to catch overcharges or missing discounts.

This manual dependency creates a cascade of further inefficiencies:

- **Estimating Bottlenecks & Capacity Constraints:** The time-intensive nature of manual drawing takeoffs means estimators spend hours counting symbols on architectural elevations and floor plans rather than strategising on pricing or negotiating better rates. This limits the volume of tenders the business can accurately process.
- **Margin Erosion through Pricing Errors:** Manually comparing supplier quotes across six different spreadsheet formats and tracking volatile material prices leaves room for human error. Furthermore, when suppliers fail to apply agreed discounts on their monthly invoices, the manual checking process often misses these discrepancies, leading to direct financial loss.
- **Labour Tracking Inaccuracies:** Processing handwritten PDF timesheets manually and cross-referencing them against the complex labour tab (which splits pricing across 10 categories like Ducting, 1st Fix, Finals, and EV Chargers) is highly prone to error. Operatives using inconsistent wording or claiming for plots that have already been paid can easily slip through, resulting in overpayments and skewed project profitability data.
- **Lack of Performance Visibility:** Because the data is locked in disparate spreadsheets, PDFs, and EQUE2, it is incredibly difficult to identify trends, such as which operatives consistently take longer on specific house types, or which plots are consistently underpriced at the tender stage.

---

3. The Solution: A Bespoke AI Commercial Assistant

We will build a bespoke AI-powered digital assistant that operates directly across your existing estimating and commercial workflows. This is not a new software platform to learn; it is an intelligent agent that performs the heavy lifting of data extraction, comparison, and tracking within the tools you already use, including Monday.com, EQUE2, Sage, and your highly structured Excel workbooks.

The automation will utilise advanced AI vision capabilities (specifically GPT-4o Vision) to read full architectural PDF packs, identify the relevant floor plans, and automatically perform component takeoffs. It will standardise the supplier quoting process, intelligently comparing returns from all six of your regular suppliers to secure the best margins. Once a project is live, the agent will use a dedicated OCR engine to process handwritten operative timesheets, allocate costs to the correct plots and fix stages, prepare the data for EQUE2, and meticulously scan supplier statements to flag overcharges or anomalies. It acts as a tireless commercial guardian, protecting your margins and freeing your human team to focus on strategic project management and negotiation.

Proof of Concept: AI Vision Takeoff

To demonstrate the capability of the AI vision model, we ran a sample architectural electrical plan through the system. This was a standard residential floor plan with BS 1192 / IEC electrical symbols. The AI successfully identified the rooms, counted the components, and produced a structured output without any human intervention or pre-configuration.

AI Output Example (Sample House Type):

Room	Recessed Downlights (R)	Ceiling Pendants	Double Sockets	Single Sockets	Light Switches	DP Switches ($)	Cooker Switch ($$)
Living Room	14	1	5	0	2	0	0
Kitchen	0	3	4	0	2	2	1
Bedroom 1	0	1	3	0	1	0	0
Bedroom 2	2	1	3	0	2	0	0
Bedroom 3	2	1	2	0	1	0	0
Study	2	1	3	0	2	1	0
WC (Main)	1	0	1	0	1	1	0
WC (En-suite)	1	0	0	0	1	0	0
Totals	**22**	**8**	**21**	**0**	**12**	**4**	**1**

Additional Items Detected:

- 1x Consumer Unit / Distribution Board (top-left corridor entry point)
- 6x Circuit wiring runs identified (colour-coded: lighting, ring main, cooker, shower, etc.)
- Smoke / heat detector positions identified at key locations

This output is exactly the data required to populate the specific house type tabs (e.g., Giganti, Crocus, Conte) in the Jeffway estimating spreadsheet. The AI reads the plan, counts the components, sections them by room, and produces a structured output that can be dropped directly into the existing workflow.

---

4. Expected Impact & Quantifiable Value

The implementation of this automation will deliver immediate, measurable, and substantial returns across your commercial operations.

Area of Impact	Benefit & Value Proposition
Time Saving & Capacity Creation	Estimators and commercial administrators will reclaim hundreds of hours per month currently spent counting symbols on 46-page drawing packs and transcribing handwritten timesheets into EQUE2. This time will be redirected to pricing more tenders and managing live projects proactively.

Margin Protection & Cost Control	Automated invoice checking ensures that every supplier overcharge, missed discount, or unauthorised purchase is immediately flagged. This directly protects project margins by preventing value leakage across the 130,000+ material lines processed annually.
Enhanced Accuracy & Consistency	Removing human data entry from the takeoff and timesheet allocation processes eliminates transcription errors, ensuring that tenders are perfectly accurate and operatives are paid correctly for the exact fix stage completed.
Performance Analytics	By digitising the timesheet and labour tracking process, the business gains instant visibility into operative performance, allowing management to identify inefficiencies and adjust future tender pricing accordingly.
Accelerated Quoting Lifecycle	The time from logging a tender to submitting a fully priced proposal will be dramatically reduced. Automated takeoffs and instant multi-supplier quote comparisons accelerate the entire estimating cycle.

---

5. Implementation Notes

Our investigation confirms that the core workflow relies on complex architectural PDF packs, handwritten PDF timesheets, and highly structured, multi-tab Excel spreadsheets. The automation will therefore be built using a two-step approach for document reading: a dedicated OCR engine (such as Google Document AI or Azure Document Intelligence) to extract the text first, then passing it to GPT-4o Vision for normalisation, spatial understanding, and logic. This two-step approach gives significantly better accuracy than vision alone, especially on handwritten timesheets or dense architectural elevations. AI vision models are very capable at reading architectural drawings and counting electrical components, and this is well-established technology (dedicated tools like Drawer AI already do this commercially with up to 70% time savings), so we are confident in the approach.

We will establish a dedicated SharePoint or OneDrive folder structure to act as the trigger point for new tender uploads. The AI will be trained to recognise your specific spreadsheet formats (including the 173-row Communals tab and the 10-category Labour Prices sheet) to ensure seamless data population.

Crucially, tasks requiring commercial negotiation or final sign-off (such as challenging a supplier on price or approving a final tender letter) are explicitly preserved as human-in-the-loop checkpoints.

Note: The investment figures detailed in Section 8 cover the design, build, and implementation of the automation. The ongoing API usage costs for the AI vision and OCR models are not included and will be billed as a running monthly operational cost based on usage volume.

---

6. Full, Unabridged Automation Breakdown

This section details every step the AI agent will perform. This is the definitive scope of work.

Module 1: AI-Powered Electrical Plan Takeoff

- **Trigger:** An estimator uploads a new tender pack (architectural PDFs and specifications) into a designated 'New Tenders' SharePoint folder, and the project is logged in the new build pipeline.
- **Step 1.1: Document Parsing & Page Identification:** The agent scans the uploaded architectural packs (often 40+ pages per house type, including substructure plans, setting out plans, joists plans, roof plans, elevations, sections, and schedules). It intelligently identifies and extracts only the relevant floor plans (e.g., 'Ground Floor Setting Out', 'First Floor Setting Out') while ignoring elevations, roof truss details, staircase details, and substructure plans that do not contain electrical data. It also reads the cover sheet to identify the house type name, the applicable plots, the specification level (e.g., Lovell Gold, Lovell Silver, Shared Ownership, Rented), and whether the build is Masonry or Timber Frame.
- **Step 1.2: Symbol Key Detection:** If the tender pack includes a symbol legend or key, the agent ingests it and maps each symbol to a component category. If no key is provided, the agent defaults to the standard BS 1192 / IEC electrical symbol library. The estimator can supply a custom key at any time, which the agent will store for future use with that developer or architect.
- **Step 1.3: AI Vision Extraction (Houses):** For standard housing plots, the agent processes the identified floor plans for each house type (e.g., Giganti, Crocus, Conte, Lunaria, Violet). Using GPT-4o Vision, it scans the plan room-by-room, counting every electrical component: single sockets, double sockets, USB sockets, fused spur units, light switches (1-gang, 2-gang, 3-gang, dimmer), ceiling pendants, recessed downlights, extractor fans, cooker switches, cooker control units, double-pole switches, consumer units, MCBs, RCBOs, smoke detectors, heat detectors, CO detectors, shaver sockets, TV plates, data plates, oven connections, hob connections, ASHP connections, EV charger points, and any PV connections. Each item is logged against the room it appears in.
- **Step 1.4: AI Vision Extraction (Flats):** For flat developments, the agent processes each flat type drawing individually (e.g., Apartment M1A GF, Apartment M1B GF, Apartment M1C FF), performing the same room-by-room component count as Step 1.3. Each flat type is treated as a distinct unit with its own takeoff. The agent also identifies the floor level (Ground Floor, First Floor, Second Floor) for each flat type.
- **Step 1.5: Communal Area Separation:** For blocks of flats, the agent identifies and separates communal areas (entrance lobbies, corridors, stairwells, landings, bin stores, cycle stores, plant rooms, retail units) from individual apartments. Communal components are logged into their own dedicated section, broken down by floor and area. This includes communal lighting (emergency and standard), small power (landlord sockets), distribution boards, submains cabling, containment (cable tray, basket, trunking), TV aerial distribution, door entry and intercom systems, fire alarm devices, and AOV (Automatic Opening Vent) systems.

- **Step 1.6: Large Block Sectioning:** For large-scale developments, the agent sections the communal takeoff into the standard Jeffway categories: Design, Prelims, Distribution, BNO Works, Containment, Small Power, and Lighting. Each section maps directly to the corresponding area in the Communals tab of the estimating spreadsheet. Within each section, the agent separates Materials (with quantities and costs) from Labour (with estimated hours and labour costs).
- **Step 1.7: Spreadsheet Population (House Type Tabs):** The agent opens the Jeffway estimating template and populates the individual house type tabs (e.g., 'Giganti', 'Crocus', 'Conte', 'Lunaria', 'Violet'). For each tab, it fills in the Qty, Manufacturer (defaulting to Deta where applicable), Part Number, Description, Unit Price (from historical data or left blank for supplier pricing), and Total columns. The structure follows the existing format: consumer units and MCBs first, then sockets, switches, light fittings, accessories, and specialist items.
- **Step 1.8: Spreadsheet Population (First Fix):** The agent populates the master '1st Fix' tab with all cable quantities (e.g., 6242Y 2.5mm, 6242Y 1.5mm, SWA), back boxes (1-gang, 2-gang, circular), fixings, and sundries. These are categorised across the columns by house type, matching the existing 47-column layout.
- **Step 1.9: Spreadsheet Population (Communals):** For flat developments, the agent populates the 'Communals' tab, filling in quantities across all subsections: Submains, Power, Low Voltage & Aerial, Lighting, AOV, Door Entry, Bin Stores, Cycle Stores, and Retail Units. It calculates both material costs and labour hours for each section.
- **Step 1.10: Accommodation Schedule Cross-Reference:** The agent reads the Accommodation Schedule provided in the tender pack (listing every plot number, house type, beds, tenure, specification, and garage type) and uses this to calculate site-wide totals. It multiplies the per-type takeoff quantities by the number of plots of each type to produce the aggregated figures for the master 'Tender' summary tab.
- **Step 1.11: Special Requirements Check:** The agent checks the specification and drawings for special requirements including EV charging (identifying which plots require it and whether it is on-dwelling or separate), solar PV connections, attached vs. detached garages (with different electrical requirements), site compound electrical requirements, and any sales area or show home specifications.
- **Step 1.12: Discrepancy Flagging:** The agent cross-references the written specification against the drawings. If a discrepancy is found (e.g., spec mandates bathroom downlights but drawings show pendants, or spec requires a specific manufacturer but drawings show a generic symbol), it generates an alert note on the summary tab for the estimator to query with the developer before finalising.
- **Step 1.13: Estimator Review Checkpoint:** The agent pauses the workflow and sends a notification to the estimator that the takeoff is complete and ready for review. The estimator reviews the quantities in the spreadsheet and provides approval to proceed to pricing. This is the critical safety net before anything feeds into the commercial stages.

Module 2: Labour Pricing & Estimation

- **Trigger:** The estimator reviews and approves the completed takeoff from Module 1.
- **Step 2.1: Historical Data Retrieval:** The agent accesses the historical pricing database, pulling labour costs from previous tenders of comparable size and specification (e.g., previous Lovell, Crest Nicholson, or Taylor Wimpey projects). It retrieves the standard labour times from the Luckins reference data for each component type.

- **Step 2.2: Labour Calculation (Houses):** For each house type, the agent calculates the base labour cost broken down by the 10 standard Jeffway categories: Ducting, 1st Fix, 2nd Fix, Finals & Heating, Fit Appliances, Weather Compensator, 20mm Condensate Pipe, Test, EV Charger, and Garages. The calculation is based on the total component count for that house type multiplied by the standard labour time, then multiplied by the agreed rate.
- **Step 2.3: Labour Calculation (Communal):** For communal areas in flat developments, the agent calculates labour costs by section (Containment, Lighting, Small Power, Distribution, AOV, Door Entry, etc.), using the agreed hourly rate (e.g., £20/hour as per the existing Communals tab) multiplied by the estimated hours per section. It produces both a per-section cost and a total communal labour package figure.
- **Step 2.4: Historical Comparison:** The agent compares the calculated labour price against actual costs from previous projects of similar size and complexity. If the calculated figure deviates significantly from historical actuals (e.g., the tender says £400 per 1st Fix for a Crocus but previous actuals averaged £450), the agent flags this for the estimator's attention with a clear recommendation.
- **Step 2.5: Markup Application:** The agent applies the standard commercial markups to generate the final tender labour price. Based on the existing Jeffway structure, this includes the Labour Markup of 26.5% (NI & Holidays), Office Overhead of 14%, and Net Profit Margin of 9.9%. These percentages are stored centrally and can be updated at any time.
- **Step 2.6: Operative Pricing Sheet Generation:** The agent generates the 'Labour Prices' sheet that will be issued to operatives on site. This lists every plot number, house type, tenure (OM/Affordable), specification (Gold/Silver/Shared Ownership/Rented), and the agreed price-work value for each of the 10 fix stage categories. This is the sheet the lads use to know what they get paid per plot per fix.
- **Step 2.7: Margin Pot Calculation:** The agent calculates the difference between the labour figure included in the tender (what the client pays) and the price-work figure issued to operatives (what the lads get paid), highlighting the retained margin per plot and per fix stage. This gives the commercial team instant visibility of the labour margin pot.
- **Step 2.8: Human Override Checkpoint:** The agent highlights the suggested labour pricing in the spreadsheet, pausing the workflow to allow the estimator to review and adjust for known complexity factors (e.g., three-storey properties, difficult access, non-standard layouts, timber frame vs. masonry differences). The estimator provides final approval before the workflow proceeds.

Module 3: Supplier Quoting & Comparison

- **Trigger:** The estimator approves the material quantities and labour pricing from Modules 1 and 2.
- **Step 3.1: Quote Request Generation:** The agent extracts the consolidated material list from the master spreadsheet and generates standardised Excel quote request forms. The forms include product descriptions, manufacturer product codes (e.g., Deta SD1209CHW for a 2G Switched Socket), and required quantities, with the unit cost column left blank for the supplier to complete.
- **Step 3.2: Supplier Selection & Dispatch:** The agent references the stored supplier list to identify the designated wholesalers for this project. While three suppliers is the standard approach, the agent can dispatch to up to six (from the pool of Templegate, Eddis & Walsall, KEW, Medlock, Moss, and Rexel) if requested. The agent automatically emails the quote requests to all selected wholesalers, requesting pricing to be returned in the provided format within a specified timeframe.
- **Step 3.3: Quote Ingestion & Format Normalisation:** As suppliers return their completed sheets (via email), the agent ingests each response. Because each supplier uses different column formats (e.g.,

Templegate uses 15 columns including DATE, CODE, DESCRIPTION, QTY, PRICE, DISC, and SELL; Eddis & Walsall uses 7 columns; KEW uses 17 columns with separate Disc% and Value fields), the agent normalises all returns into a single, unified comparison structure. It validates that every line item has been priced and flags any gaps or items marked as 'not available'.

- **Step 3.4: Line-by-Line Comparison:** The agent compares the unit prices line-by-line across all returned quotes. It produces a comparison matrix showing each item, the quoted prices from each supplier, and the percentage variance between them.
- **Step 3.5: Rebate & Net Cost Calculation:** The agent factors in any known annual or quarterly rebates for each supplier (stored in the system and updated periodically). It calculates the true net cost per supplier after rebates are applied, which may change the ranking.
- **Step 3.6: Cable Price Volatility Check:** For cable items specifically (which are known to fluctuate significantly), the agent compares the quoted prices against the last three months of cable pricing data from previous orders. If prices have moved significantly, it flags this to the estimator with a recommendation on whether to lock in now or wait.
- **Step 3.7: Best Package Recommendation:** The agent identifies the most cost-effective supplier for the overall package and presents a summary showing the total cost from each supplier (pre- and post-rebate), along with a clear recommendation.
- **Step 3.8: Human-in-the-Loop Negotiation Flag:** The agent highlights specific line items where a non-selected supplier is significantly cheaper (e.g., light switches at £1 from Supplier B vs. £3 from the recommended Supplier A). It generates a negotiation brief for the commercial team, suggesting they approach the preferred supplier to match on those specific items before confirming the order.
- **Step 3.9: Specialist Item Routing:** If certain items (e.g., specialist light fittings, solar PV mounting kits, specific manufacturer products) are not stocked by the main wholesalers or are significantly cheaper direct from the manufacturer, the agent flags these for separate procurement and excludes them from the main package comparison. It can also route specialist items to sub-contractors for pricing where required.
- **Step 3.10: Final Population & Handover:** Upon final supplier selection by the commercial team, the agent populates the chosen unit prices into the master tender spreadsheet. It also generates the second fix takeoff sheets per plot type that will be sent to the chosen supplier for order fulfilment once the project is won. The agent advises finance/commercial of the agreed costs against the takeoffs.

Module 4: Tender Document Generation

- **Trigger:** The master tender spreadsheet is fully populated and the estimator marks the tender as 'Ready for Submission'.
- **Step 4.1: Data Extraction:** The agent extracts the final pricing, quantities, plot types, material specifications, supplier details, and labour allowances from the completed spreadsheet, including the Accommodation Schedule data (plot numbers, house types, beds, tenure, specification, garage types).
- **Step 4.2: Tender Letter Drafting:** The agent opens the standard Jeffway Tender Letter Word template and maps the variables correctly. It populates the document with the specific project details, including a section-by-section breakdown of allowances (Distribution, Containment, Lighting, Small Power, etc.), the agreed rates, material markups, and total costs per house type and for communal

areas. The letter is drafted as a professional scope document that clearly communicates the value of the proposal.

- **Step 4.3: Inclusions Section:** The agent drafts a detailed inclusions section listing everything Jeffway is allowing for in the tender: specific consumer unit types, socket specifications (e.g., Deta Slimline range), light fitting types, cable specifications, testing and certification, EV charging (if applicable), PV connections (if applicable), ASHP connections, and any specialist installations. This section is comprehensive enough that there is no ambiguity about what is covered.
- **Step 4.4: Exclusions & Qualifications:** Based on the discrepancies flagged in Step 1.12 and standard company policy, the agent drafts the specific exclusions (items not included in the price) and qualifications (assumptions made, e.g., 'quotation based on specification Rev D dated [date]'). This protects Jeffway commercially by making the scope crystal clear.
- **Step 4.5: Tender Summary Sheet:** The agent ensures the final 'Tender' summary tab in the spreadsheet is perfectly formatted for client submission, showing house types, plot counts, total value, cost per plot (with markups), materials per plot, and labour per plot.
- **Step 4.6: PDF Conversion & Storage:** The agent saves the drafted Word document, converts a copy to PDF, and stores both alongside the final spreadsheet in the project's SharePoint folder, labelled as Revision A.
- **Step 4.7: Revision Management:** When the developer requests changes (e.g., 'you're too expensive, can you do it for less?'), the estimator instructs the agent to create a new revision. The agent duplicates the spreadsheet and tender letter, applies the requested adjustments (tweaked labour rates, updated supplier buy-in, revised markups), and saves as Revision B, C, D, etc. Each revision is stored alongside the previous versions for a full audit trail.
- **Step 4.8: Submission Notification:** The agent notifies the estimator that the tender pack (spreadsheet + letter + PDF) is ready for submission, with a link to the SharePoint folder.

Module 5: Timesheet Processing & Labour Tracking

- **Trigger:** An operative emails their weekly handwritten PDF timesheet to the Contracts Manager, who checks and approves it before forwarding to the finance inbox.
- **Step 5.1: OCR Data Extraction:** The agent uses a dedicated OCR engine (such as Google Document AI or Azure Document Intelligence) to read the handwritten PDF timesheet. It extracts the operative's name, employee reference (e.g., JWE262), site code (e.g., 2168 Ph6), plot numbers, descriptions of work, and the hours or price-work amount claimed for each line item. The two-step OCR-then-GPT approach gives significantly better accuracy than vision alone on handwritten or inconsistently formatted timesheets.
- **Step 5.2: Terminology Normalisation:** The agent passes the OCR text to GPT-4o to normalise inconsistent wording. For example, 'PL2' is mapped to 'Plot 2', 'finals' is mapped to 'Finals & Heating', and abbreviations are resolved against the project's known 10 fix types (Ducting, 1st Fix, 2nd Fix, Finals & Heating, Fit Appliances, Weather Compensator, 20mm Condensate Pipe, Test, EV Charger, Garages).
- **Step 5.3: Site Code Validation:** The agent verifies that the site code on the timesheet matches a live project in the system (cross-referencing against the 'Current Works' tab in the Wages Summary). If the site code is unrecognised, the line item is flagged for manual review.
- **Step 5.4: Plot & Fix Availability Check:** The agent checks the master 'LABOUR' tab for the relevant project to confirm that the specific plot and fix type have not already been claimed by another

operative. If the cell is already populated with another operative's initials, the agent flags this as a potential duplicate claim and generates an alert.

- **Step 5.5: Amount Verification:** The agent compares the amount the operative is claiming against the agreed price-work value for that fix on that plot type (from the Labour Prices sheet generated in Module 2). If the operative is claiming more than the allowed amount, the agent flags the discrepancy. If they are claiming less (e.g., partial completion), the agent logs it as a partial claim.
- **Step 5.6: Day Work Handling:** For operatives working on hourly rates or day rates (common on larger communal area installations and for apprentices), the agent extracts the hours claimed and verifies there is an accompanying daywork sheet reference. If no daywork reference exists for hourly work, the agent flags the item to be reverted back to the operative. Validated day work is logged against the communal labour budget rather than the plot-based labour tab.
- **Step 5.7: Discrepancy Report:** For any line items that fail validation (duplicate claims, overcharges, unrecognised site codes, amount mismatches, missing daywork references), the agent compiles a discrepancy report and emails it to the finance team. Each flagged item includes the operative's name, employee reference, the claim details, and the specific reason for rejection.
- **Step 5.8: Labour Tab Population:** For all validated claims, the agent inputs the operative's initials and the claimed amount into the correct cell on the master 'LABOUR' tab for the relevant project application spreadsheet. This automatically feeds into the main application sheet (as the labour tab is linked to the application).
- **Step 5.9: Wages Summary Update:** The agent simultaneously updates the individual operative's tab on the master JWE Wages Summary spreadsheet (e.g., the 'Alex Hanley' tab, the 'Andy Bond' tab). It enters the site codes, plot numbers, and amounts under the correct week-ending date columns, maintaining the existing 81-row x 22-column structure per operative.
- **Step 5.10: Summary Tab Aggregation:** The agent updates the master 'Summary' tab on the Wages Summary, ensuring the Total Gross, Standard Hours, Price Work, Holiday/ROP, Bank Holiday pay, Phone, Van, and Mileage columns are all correctly aggregated for each operative. It also updates the 'Timesheet Checklist' tab to confirm receipt and processing.
- **Step 5.11: EQUE2 Preparation:** The agent compiles all validated timesheet data into the exact format required for direct entry into the EQUE2 payroll system, which feeds through to Sage Payroll. This includes the gross pay breakdown, salary sacrifice pension deductions, and the net payable amount per operative.
- **Step 5.12: Performance Analytics:** The agent continuously logs completion data against historical averages. It flags management if specific operatives are consistently taking longer on comparable plots or house types (e.g., if Operative A is claiming 2nd Fix on a Crocus in 2 days but the average is 1.5 days). It also flags if certain plot types are proving more expensive in labour than the tender allowed. This data feeds back into Module 2 for future tender pricing accuracy.
- **Step 5.13: Underpayment & Overpayment Alerts:** If the cumulative labour spend on a project is tracking significantly above or below the tendered labour allowance, the agent sends a proactive alert to the commercial team, enabling early intervention rather than discovering the issue at project completion.

Module 6: Material Spend Tracking & Discrepancy Detection

- **Trigger:** A supplier emails their monthly Excel statement or invoice breakdown to the finance inbox, or the file is uploaded to the designated SharePoint folder.

- **Step 6.1: Statement Ingestion & Normalisation:** The agent ingests the Excel statement from the supplier using the same OCR-first approach as Module 5. Because each of the six suppliers uses different column formats (e.g., Templegate uses 15 columns with DATE, NUMBER, CODE, DESCRIPTION, QTY, PRICE, PER, DISC, SELL, SELLING, and ORDER REF; Eddis & Walsall uses 7 columns; KEW uses 17 columns with separate Disc% and Value fields; Medlock uses 9 columns), the agent normalises all formats into a standard structure. We will need sample statements from each supplier upfront to map the column structures before building the ingestion logic.
- **Step 6.2: Order Reference Parsing & Normalisation:** The agent reads the order reference on each line item (e.g., '2145 11 LLANYRNEWYDD', '2226/STONEWOOD RETURN', 'TW NAILSEA', '2168 PV') and maps it to the correct project site code, plot number, and fix type. Where operatives or site foremen have used inconsistent formatting, the agent applies normalisation logic to resolve the reference.
- **Step 6.3: Project Spend Allocation:** The agent allocates each line item to the correct project in the master Project Spend Tracker. It logs the material cost against the specific project, building a cumulative spend picture.
- **Step 6.4: Price Verification:** The agent cross-references the charged unit price on each line item against the agreed tender price established with the chosen supplier in Module 3. If the supplier has charged a higher price than agreed (e.g., due to an unapproved price increase), the agent flags the item with the exact financial impact calculated.
- **Step 6.5: Discount Verification:** The agent checks that the correct discount percentage has been applied to each line item. Suppliers sometimes fail to apply the agreed discount, resulting in overcharges. For example, if KEW has agreed a 70% discount on cable glands but only applied 65%, the agent flags the difference and calculates the financial impact.
- **Step 6.6: Quantity Monitoring:** The agent compares the cumulative quantity of each item purchased for the project against the total quantity allowed in the original takeoff. If the site has ordered 150 double sockets but the takeoff only allowed for 120, the agent flags the overrun and calculates the cost of the excess.
- **Step 6.7: Unrecognised Purchase Detection:** The agent identifies any line items with order references that do not match a known project or plot. These are flagged as potential unauthorised purchases (e.g., operatives buying materials for personal use on the company account). The agent generates a separate alert for these items.
- **Step 6.8: Returns & Credits Processing:** The agent identifies negative quantities (returns) in the supplier statements and reconciles them to ensure the project spend is accurately reduced. It tracks handling charges on returns and ensures they are accounted for.
- **Step 6.9: Monthly Exception Report:** The agent generates a comprehensive monthly exception report for the finance team, organised by project. The report includes: items charged above the agreed rate, missing or incorrect discounts (with the exact financial impact), quantity overruns by item, unrecognised or suspicious purchases, and a project-level summary comparing total material spend to date against the tendered material allowance.
- **Step 6.10: Supplier Credit Note Tracking:** When the finance team reverts back to the supplier for a credit based on the exception report, the agent tracks the outstanding credit notes and reconciles them against subsequent statements to ensure the credits are actually applied.
- **Step 6.11: Profitability Dashboard Update:** The agent updates the project profitability view, combining the material spend data from this module with the labour spend data from Module 5,

giving the commercial team a real-time view of actual costs vs. tendered allowances for every live project.

Module 7: Application Template Generation & Monthly Invoicing

- **Trigger:** A project is confirmed as won and marked as 'Awarded' in the system.
- **Step 7.1: Application Template Creation:** The agent takes the final tender revision spreadsheet and automatically generates the Main Application spreadsheet (e.g., 'JWES Main Contract Application'). It creates the SUMMARY tab with the Account Summary, the JWES Application tab with the formal interim application letter format, and the Main Application tab populated with every plot number, house type, and the agreed value per fix stage (1st Fix, 2nd Fix, Finals, Garages, Car Charging).
- **Step 7.2: Fix Stage Percentage Setup:** The agent sets up the tracking columns for each fix stage, configuring the spreadsheet to track percentage completion (0 to 1) and monetary value claimed to date. The percentage split between fix stages is configurable and stored centrally.
- **Step 7.3: Communal Area Setup:** For flat developments, the agent sets up the communal area section below the flat plots, with percentage-based claiming for each communal work package (Containment, Lighting, Small Power, Distribution, AOV, Door Entry, etc.).
- **Step 7.4: Variations Tab Setup:** The agent creates the 'Variations to Contract' tab, pre-formatted with columns for Item number, Plot, Description, Order Number/Ref, Works Value, % Complete, Claimed to Date, Paid, Outstanding, and Notes. This is ready to receive variation entries as they arise during the project.
- **Step 7.5: Labour Tab Setup:** The agent creates the 'LABOUR' tab, pre-populated with every plot number and fix stage from the tender, ready to receive operative timesheet data from Module 5.
- **Step 7.6: Monthly Template Duplication:** At the start of each application period, the agent automatically creates the new month's application (e.g., moving from Application 19 to Application 20). It copies the 'Gross Claimed to Date' figures into the 'Previous Claim' column, resetting the current month's claim to zero, ready for the contracts manager to update.
- **Step 7.7: Contracts Manager Input Workflow:** The agent sends the current month's template to the relevant contracts manager(s), requesting them to update the percentage complete for each plot and fix stage. When the contracts manager returns the updated sheet, the agent ingests the highlighted plots and updated percentages.
- **Step 7.8: Claim Calculation:** The agent calculates the current month's claim value based on the newly updated completion percentages. It produces the Gross Valuation, less Previous Valuations, to arrive at the Valuation This Month figure. It also calculates the Net Balance Due after deducting amounts already paid.
- **Step 7.9: Variation Logging:** As new variations arise during the project (e.g., customer extras like additional USB sockets at £54 per plot, drawing error corrections backed by daywork sheets like 'DW 5165', return visits, or compound works), the agent logs each one into the 'Variations to Contract' tab with the description, agreed value, order reference, and claim status.
- **Step 7.10: Client-Specific Timing:** The agent manages different application submission dates per client. For clients requiring actuals (e.g., 28th of the month), it only includes confirmed completions. For clients requiring projections (e.g., 19th with month-end valuation), it includes confirmed completions plus projected works based on the contracts manager's forecast. It also tracks the 'Final Date for Payment' for each application.

- **Step 7.11: Invoice Generation:** The agent generates the formal Interim Application letter (the 'JWES Application' tab), populating the FAO, project name, application number, application date, period ending date, order number, Forecast Final Account, Contract Works value, Extras/Variations value, Gross Application, Less Paid to Date, and Net Balance Due. This is formatted ready for submission to the client.

---

7. Training and Handover

Upon completion of the build, we will conduct a full training and handover session. This is not about learning a new system, but about learning how to work with your new digital commercial assistant. We will cover:

- How to structure and upload architectural drawing packs into the SharePoint folder to ensure the AI vision model extracts data perfectly, including best practices for file naming and folder organisation.
- How to review and approve the AI's takeoff output before it feeds into the pricing stages, including how to spot-check quantities and override where needed.
- How to manage the human-in-the-loop checkpoints, specifically reviewing the AI's suggested labour pricing and using the supplier negotiation flags to secure better deals.
- How to review the automated discrepancy reports for handwritten timesheets and supplier invoices, and how to action the exception items efficiently.
- Best practices for standardising operative timesheet submissions to ensure maximum OCR processing accuracy, and how the data flows through to EQUE2 and Sage Payroll.
- How to use the project profitability dashboard to monitor live projects against tendered allowances in real time.
- How the revision management system works for tender resubmissions and how to instruct the agent to create new revisions.
- How to update the stored parameters (markup percentages, supplier rebate rates, labour times) as the business evolves.